

Use case	Python libraries	Business applications
Customer segmentation	scikit-learn, pandas, PyClustering	Targeted marketing, personalised recommendations
Image segmentation	scikit-image, OpenCV, PyClustering	Object detection, image compression, medical image analysis
Anomaly detection	scikit-learn, PyOD, PyClustering	Fraud detection, network security, quality control
Document clustering	NLTK, Gensim, scikit-learn	Document organisation, information retrieval, summarisation
Genomics and bioinformatics	Biopython, scikit-learn	Understanding genetic relationships, identifying disease biomarkers, drug discovery
Market segmentation	scikit-learn, pandas, PyClustering	Tailoring products, pricing, marketing strategies
Pattern recognition	scikit-learn, TensorFlow, PyClustering	Pattern recognition, signal processing, data mining

Table 1: Use cases of cluster analysis

iteratively assigns data points to the nearest cluster centroid and then recalculates the centroids based on the mean of the data points in each cluster. This process continues until convergence is achieved, resulting in well-defined clusters.

For instance, let's say we have a dataset of 100 data points representing the annual income and spending score of customers in a mall. Using K-means clustering with $K=3$, the algorithm aims to partition these data points into three clusters in such a way that the sum of squared distances between each data point and its assigned cluster centroid is minimised. After several iterations, the algorithm converges, and we obtain three well-defined clusters. Customers in Cluster 1 might have moderate incomes and high spending scores, while those in Cluster 2 might have lower incomes and lower spending scores. Cluster 3 could represent customers with high incomes but moderate spending scores. This segmentation helps mall management understand different customer segments and tailor marketing strategies accordingly.

K-means algorithm has certain limitations as well. For instance, it might not always find the best solution, leading to less accurate clusters. Additionally, the algorithm requires the number of clusters (K) to be specified

beforehand, which can be challenging in some cases. Moreover, K-means is sensitive to the initial placement of cluster centroids, and changing its starting points can lead to different final groupings. Despite these limitations, K-means remains a widely used and efficient algorithm for partitioning data into clusters, offering valuable insights into the underlying structure of datasets in various fields ranging from customer segmentation to image processing.

Minimum data and the number of clusters required

The minimum amount of data required for cluster analysis depends on various factors such as the complexity of the dataset, the number of features, and the desired level of granularity in the clustering results. If the number of data points is too small, the clusters may be unstable, and the clustering results may not be reliable. In general, one would typically need a dataset with at least a few dozen data points to capture meaningful patterns and relationships within the data to perform basic cluster analysis effectively. However, for more complex datasets or when dealing with high-dimensional data (like images with many pixels), hundreds or even thousands of data points may be required to obtain reliable clustering results. For instance, in image segmentation tasks

where each pixel constitutes a data point, an image with 1000×1000 pixels would require at least a million data points for clustering to be meaningful.

The number of clusters chosen for analysis is also crucial. In market segmentation scenarios, where consumer behaviour is diverse, a maximum of 8-10 clusters may be appropriate to capture the different market segments effectively. Conversely, in crop classification tasks where the variety of crops is limited, a maximum of 4-6 clusters may suffice to categorise the crops accurately. Having enough data points per cluster is also essential to ensure stable and meaningful clustering. For instance, in financial risk assessment, each risk profile cluster should ideally contain a minimum of 100 data points to provide reliable insights.

In summary, while there is no fixed minimum threshold for the amount of data required for cluster analysis, it's crucial to consider the dataset's characteristics and the desired level of granularity to determine the appropriate number of data points and clusters accurately.

Clustering example based on synthetic data

Understanding customer preferences is crucial for businesses. In this section, we delve into the implementation of cluster analysis using Python to